Passive Friction Measurement Pilot in Ghana's Digital Payment Corridors

Technical Annex, shared by all cover notes

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1. Summary

The request is for cooperation to run a 30 day, read only measurement of settlement delay and friction in a defined cluster of payment infrastructure. The measurement runs inside the operator's own environment under a pre registered protocol. No money moves, and no existing system changes.

The outputs are a baseline friction report for the operator and, with the operator's consent, aggregate public data. Later phases are conditional and are not part of this request.

The work is led by Kevin Ansong, Independent Researcher. It is open source, and every result will be published whatever it shows.

1.1 What is proposed: one open standard and one registered test

Operators already measure delay inside their own systems. This request adds two things.

- **An open measurement standard.** Every transaction is split into the same lifecycle stages: initiation, authentication, routing and transit, clearing wait, and settlement confirmation. Latency quantiles (median, 90th and 99th percentile), retries, failures and queue load are computed per stage and end to end in 10 minute windows. Two entropy figures summarise how the timings are spread: the entropy of the latency distribution, and the entropy of where in flight time is spent across stages (this second figure is reserved for Phase 1 operator logs, and the reference code reports a different, descriptive stage entropy; see Section 3.1). The definition is written down so that any operator, the regulator or an independent engineer can compute the same figures from the same logs. Section 3 gives the data minimisation rules.
- **A registered test.** The pre registration (OSF, https://osf.io/w6uan/) tests whether the entropy of the latency distribution predicts the 99th percentile latency of the next window beyond a gradient boosted model that already uses lagged latency and queueing features. The rules for support, failure and inconclusive results were fixed before any data was opened. The first test runs on public datasets. The stage entropy is reserved for operator logs in Phase 1.

Stage timing and percentile latency are ordinary engineering practice. What is proposed here is a shared written definition, the entropy figures, and the privacy rules, packaged for comparison across operators. The standard has no standing until operators or the regulator adopt it, and the predictive value of the entropy figures is not yet known.

2. Background and status of the evidence

The Sovereign Negentropy working paper (October 2026) proposes a voluntary layer that runs alongside existing payment and administrative systems. The paper defines a friction entropy index, a demurrage currency with a utility offset, and a velocity dependent fee.

What exists today:

- The paper has not been independently reviewed.
- Its mathematics has been checked by 47 offline programmatic checks (28 numerical and 19 symbolic). They test internal consistency only. They do not test whether the economics works.
- Every empirical claim in the paper is a hypothesis. None has been tested.
- A pre registered protocol (version 1.1) fixes the first tests and states what result would end the entropy claim.

Phase 1 does not depend on the paper being right. It produces measurements of settlement delay, retries, failures and queue behaviour that are useful to the operator whatever the index shows.

3. Phase 1: passive friction measurement

3.1 What is measured

- Lifecycle stage durations for each transaction: initiation, authentication, routing and transit, clearing wait, settlement confirmation.
- Latency quantiles (median, 90th and 99th percentile) per stage and end to end, by 10 minute window.
- Retry counts, webhook retransmissions, error and failure rates.
- Queue utilisation estimates and arrival rates.
- Fees charged per transaction band.
- The entropy of the latency distribution, and a descriptive entropy of the stage durations booked at stage exit. The in flight occupancy statistic defined in the pre registration is not yet implemented in the reference code. The friction terms are defined in the pre registration.
- Pipeline overhead on the host (processor, memory, input and output), measured throughout.

3.2 Data minimisation

Item	Treatment
Event type and timestamp	Required
Transaction identifier	Required, replaced by a salted hash inside the operator environment
Channel or rail, status code, retry count, fee amount	Required
Amount	Banded (value bands agreed with the operator) or exact, at the operator's choice
Phone number, name, account number, KYC data, location	Excluded. Never read by the pipeline
Output cells	Aggregates only. Any window, and any stage figure within a window, with fewer than 50 contributing transactions is suppressed

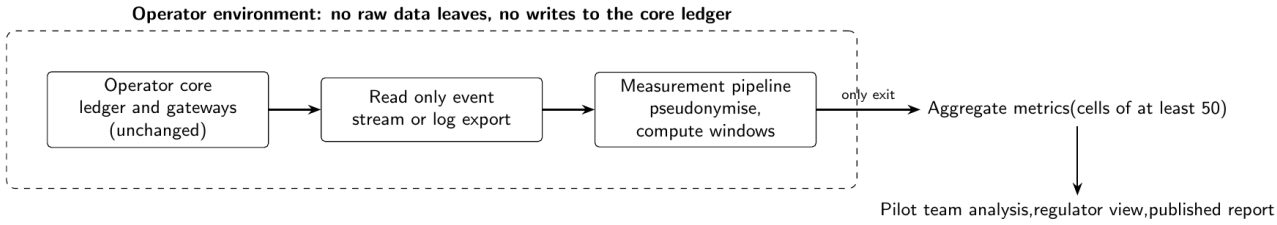


Figure 1: Phase 1 data flow. Raw data stay inside the operator environment. Only aggregate metrics leave.

3.3 Deployment pattern

- The pipeline reads from a replica, an existing event stream or a scheduled log export. It performs no writes to the core ledger and takes no write locks on production tables. Its read queries take ordinary read locks only.
- It runs on hardware designated by the operator. The operator controls its credentials and can revoke them at any time.
- Releases are signed. The operator reviews the source code and the dependency list before deployment and may run its own build.
- A test run on synthetic data precedes any run on operator data.

3.4 Outputs and publication terms

- A baseline friction report for the operator, delivered first.
- An analysis under the Phase 1 pre registration, filed before collection starts.
- With the operator's consent, a public release of aggregate metrics.
- The operator may review the report for factual accuracy and confidentiality for 30 days before publication. The operator has no veto over the findings.
- Results are published whatever they show.

3.5 Resource limits and stop rules for Phase 1

- The operator sets a cap on the pipeline's resource use at the technical session. The pipeline stops itself if the cap is exceeded.
- The operator or the regulator may stop the pipeline at any time without notice or reason.
- Any suspected security incident halts collection and preserves evidence.

3.6 Validation of the measurement path

The repository contains a reference pipeline in the telemetry folder, written in the Python standard library, with three tests that anyone can rerun.

- Correctness: 22 of 22 self checks pass on synthetic data with known answers. The streaming histograms equal a batch computation exactly, stage quantiles recover the known lognormal truth within 8 percent, retry and failure totals equal the injected totals, windows and individual stage figures under 50 contributing transactions are suppressed, duplicate and replayed transaction identifiers are rejected, abandoned transactions expire (including at the end of a batch or stream, through flush and advance_watermark) and old windows are evicted so memory stays bounded, rejected events never move the expiration clock, durations above the histogram range are counted

and flagged, and the source imports only the standard library. The checks test the reference code. They are not a privacy or security review.
- Cost per event: 0.95 to 1.35 microseconds per event from 50,000 to 3,200,000 events, so the cost does not grow with history. Memory is bounded by transactions in flight.
- Locking: on PostgreSQL 16 with four writers at 800 transactions per second for 30 seconds, five repetitions each, the read only snapshot reader caused no lock waits beyond those between writers, and no deadlocks. The few waits recorded were writers meeting each other, at the same level with and without the pipeline. The 99th percentile write latency was 3.33 ms with the pipeline and 3.20 ms without it (medians over five runs, pipeline version 0.2). A control reader that deliberately held a table lock raised the 99th percentile to about 160 ms, which shows the test detects locking when it occurs.

Limits: the load is synthetic and ran on one shared two core machine. These tests show the design takes no write locks and has a constant per event cost. They do not predict the behaviour of any operator's production system, so the first task of Phase 1 is to repeat the locking test on the operator's own replica.

Review record: the telemetry code was reviewed on 4 November 2026 by an external reviewer. The review reproduced eight weaknesses, which were fixed or relabelled on 5 November 2026. A follow up review on 9 November 2026 confirmed those fixes and found two further issues, fixed in pipeline version 0.3: stale transactions are now expired at the end of a batch or stream, and rejected events no longer move the expiration clock. REVIEWERS.md in the repository lists each finding and its resolution. Known limits that remain: the stage figures in one window describe different cohorts of transactions and must not be added; retry and failure counts are counts, not rates; the in flight occupancy statistic of the pre registration is not implemented; read queries take ordinary read locks, so the claim is that the pipeline takes no write locks and, in the test, did not slow writers; the benchmark reader advances by sequence number, and a late committing row can be skipped, which the operator's own adapter must handle.

3.7 Indicative timeline

Stage	Indicative duration
Approvals, data protection impact assessment, security review	Set by the operator and the regulator
Test run on synthetic data	2 weeks
Collection on the agreed cluster	30 days
Analysis and operator report	15 days
Factual review and publication	30 days

4. Conditional later phases (not requested now)

Nothing in this section begins without separate written approval from the operator and the regulator, a separate registration, and the Phase 1 report.

4.1 Phase 2: opt in sandbox cohort

- A small, closed cohort of volunteer merchants.
- Participant balances are backed by funds held in ring fenced accounts at a licensed institution. The pilot team does not hold funds.
- Demurrage, utility and fee parameters are set and approved by the regulator before launch.

- Conversion to legacy currency at published cost is available at all times.
- Entry requires a successful drill of the stop procedure (Section 4.3).

Backing invariant: at every reconciliation, the value held in ring fenced accounts is at least equal to the sum of participant balances plus accrued fees and pending redemptions. The invariant accounts for demurrage and fees. It does not require exact equality.

4.2 Phase 3: corridor expansion

Expansion by corridor, one at a time, under the same stop procedure, with capacity planned from measured throughput limits.

4.3 Stop conditions

Thresholds marked Phase 1 are set from the Phase 1 baseline and approved by the regulator. No fixed number is chosen before that baseline exists.

Trigger	Threshold	Response
Tail latency	99th percentile of the parallel layer exceeds the Phase 1 baseline by a multiple set by the regulator, for a set number of consecutive windows	Pause new entries, alert, human review
Backing shortfall	Backing below participant balances plus accrued fees at any reconciliation	Freeze issuance, alert the regulator, reconcile. Redemptions continue at published cost
Queue saturation	Utilisation above a level set from Phase 1	Throttle new connections
Off ramp delay	Conversion time above a level agreed with the licensed institution	Pause new entries and alert. Any liquidity action needs human approval
Volume loss	Cohort transaction volume falls below a threshold relative to a matched control	Review. Threshold informed by the levy elasticity estimate
Security incident	Any confirmed or suspected compromise	Halt and preserve evidence

Stop authority sits with named officers of the regulator and the operator. The target time to halt is agreed with them and verified in a drill before Phase 2.

Residual risks that remain after these controls: conversion cost and delay during a rollback, operational error, losses at the licensed institution, participant confusion, and regulatory change. The pilot reports each one.

5. Security, privacy and compliance

- Data protection. The operator is the data controller. The pilot team processes data only under a written agreement. A data protection impact assessment precedes deployment. Compliance is with the Data Protection Act 2012 (Act 843), including any registration with the Data Protection Commission, as advised by counsel. No personal data leave the operator environment in Phase 1.
- Payment regulation. Any phase that holds or moves value follows the Payment Systems and Services Act 2019 (Act 987) and the Bank of Ghana sandbox rules, with funds held by a licensed institution.

- Anti money laundering and identity. Participants are identified through existing KYC at the operator and through the national identity card. The pilot team holds no KYC data and builds no identity layer.
- Threat model.

Threat	Control
Tampering with pipeline code	Signed releases, operator code review, reproducible builds
Data exfiltration	No raw data egress, aggregate output only, small cell suppression
Insider misuse	Operator controlled credentials, access logging, least privilege
Supply chain compromise	Pinned dependencies, published dependency list, operator build option
Availability impact on production	Read only access, replica or log based source, resource cap, self stop
Model risk	Pre registration, independent evaluation, publication of failures

6. Evaluation and integrity

- The first tests are fixed in the pre registration (version 1.1). They state what counts as support, failure and an inconclusive result.
- An evaluator independent of the author reviews the Phase 1 results. The evaluator is to be designated through a University of Ghana or external peer review track.
- Code, definitions and aggregate results are published in the open repository. Deviations from the registration are logged and labelled.
- If the entropy predictor is not supported, the paper is revised to say so within 30 days of the analysis.

7. Team and governance

Role	Name	Status
Principal investigator	Kevin Ansong	Confirmed
Legal entity	Sovereign Negentropy Initiative (Project Specific SPV / Pending Registration)	Pending registration
Technical lead	To Be Designated (TBD) upon sandbox phase authorization	To be designated
Quantitative lead	To Be Designated (TBD) upon sandbox phase authorization	To be designated
Regulatory and legal counsel	To Be Designated (TBD) upon sandbox phase authorization	To be designated
Independent academic evaluator	To Be Designated (University of Ghana / External Peer Review Track)	To be designated
Protocol team contact	Kevin Ansong, kevinansong101@gmail.com	Confirmed

- Any change to the protocol after approval needs written agreement from the operator and the regulator.

- Declaration of interests and funding: the work has no external funding as at 2 November 2026. The author is the founder of LastStop, a web based vendor directory, and holds no interest in any payment operator named in this document.
- Licences: code is released under the MIT licence and the paper under CC BY 4.0.
- Tools and assistance. This work was developed by Kevin Ansong with AI assistance (Claude, by Anthropic) for drafting, code and test construction. Kevin Ansong directs the project and is responsible for every claim. All code, tests and results are open and can be rerun by anyone.

Appendix A. Status of claims

Claim	Status	Basis
The paper's mathematics is internally consistent	Checked	47 offline checks (28 numerical, 19 symbolic). They do not test the economics
The friction entropy index predicts tail latency	Hypothesis	Pre registered test S1. No result yet
Demurrage with a utility offset retains balances	Hypothesis	Retention condition proved under stated assumptions (Proposition 4). Empirical retention untested
A velocity dependent fee raises velocity without lowering volume	Hypothesis	Untested. The levy elasticity estimate (S2) covers a flat fee only
A 0.05 percent fee funds 20 percent of GDP	Arithmetic	Requires taxed turnover of 400 times GDP (Proposition 9). Not a proposal to replace tax revenue
The Phase 1 pipeline adds no writes to the core ledger	Design requirement	Verified by operator code review and measured overhead during Phase 1
Phase 1 carries no monetary risk	Design property	No value is issued, held or moved in Phase 1
Zero knowledge verification takes about 1 millisecond	External figure	Order of magnitude from a third party benchmark. Not reproduced. Not used in Phase 1
The parallel layer is compatible with Ghanaian law	To be confirmed	Counsel and the regulator decide
Later phases are safe to run	Not established	Conditional on Phase 1, separate registration and separate approval